

A Pyrrhic Victory? Bank Bailouts and Sovereign Credit Risk

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August 2011

Questions

- ❶ Did financial sector bailouts ignite sovereign credit risk in the developed economies?
 - were there important immediate costs to the bailouts (as opposed to just distortions of future incentives)
- ❷ What mechanisms underlie the relationship between financial sector and sovereign credit risk?
 - transmission of risks (spillover) between the sectors
 - trade-off between financial sector and sovereign credit risk
- ❸ Does sovereign credit risk also feedback onto financial sector credit risk?
 - the ongoing banking crisis: impact of default risk in Greece, Ireland, Portugal, Italy(!)

Motivation: Bailout of Irish Banks

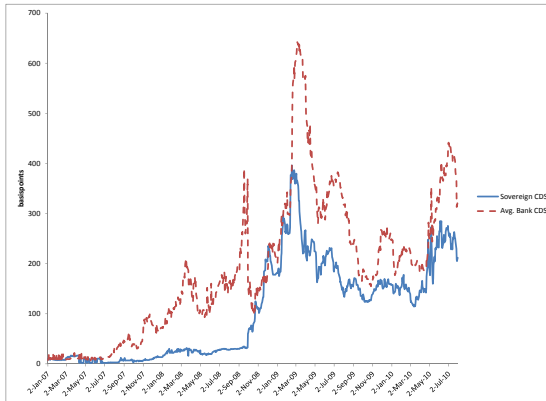
From Financial Sector Credit Risk to Sovereign Credit Risk

- On September 30, 2008 the government of Ireland announced a guarantee of all deposits of its six biggest banks
- Credit default swap (CDS) fee for buying protection on Irish banks fell from 400 bps to 150 bps
- From the standpoint of stabilizing the financial sector, the end goal of the guarantees appeared to have been met
- What impact would these provisions have on the credit risk of the government of Ireland?

Bailouts and Risk Transfer

- Just one of the Irish banks, Anglo Irish, cost the government Euro 25 Billion or 11.26% of GDP by Aug'10
- Ireland received 85 Billion Euro rescue package by European Union and IMF in Nov'10 and now needs another 24 Billion Euro for lenders
- Total is approximately 70% of 2010 GDP

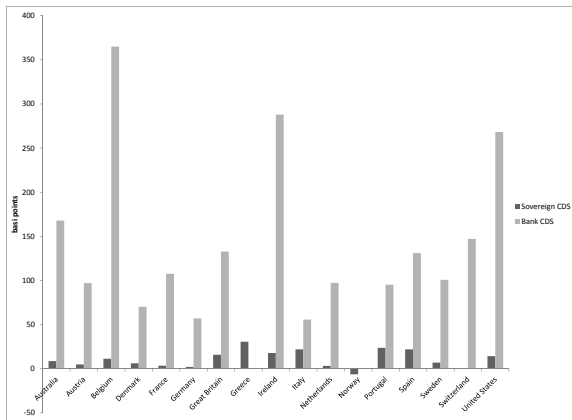
A Motivating Example: The Case of Ireland



- Chart similar across many countries:

- 1 sovereign CDS close to 0 through first-half 2008
- 2 post bailout announcement (9/30/2008): sovereign CDS jumps up, bank CDS drops down
- 3 subsequent positive comovement

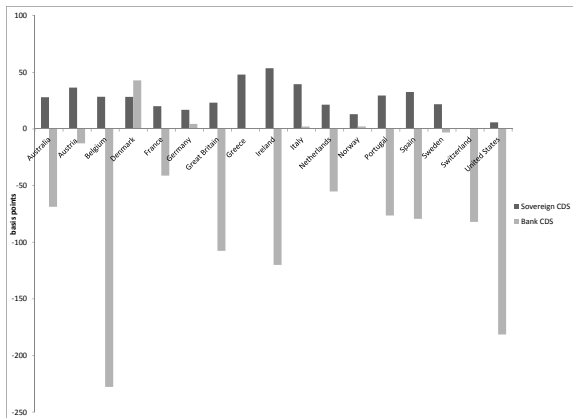
Pre-Bailouts: Europe



3/1/2007 – 9/26/2008

- bank CDS has increased substantially
- not much change in sovereign CDS

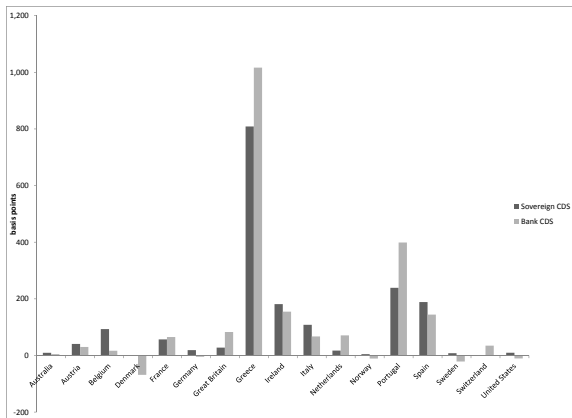
During the Bailout Period



9/27/2008 – 10/21/2008

- bank CDS decreases substantially
- strong increase in sovereign CDS

Post Bailout



10/22/2008 – 6/30/2010

- positive comovement
- a merger of financial sector and and sovereign?

This Paper

- Models trade-off between sovereign and financial sector credit risk
- Government can transfer resources to financial sector
 - Transfer alleviates under-provision of financial services (debt overhang)
 - Funding the transfer induces underinvestment in corporate sector and dilutes existing sovereign bondholders
- Solve government's problem and resulting sovereign bond price
 - 1 Under certainty about future output and no-default
 - 2 Allowing for strategic default
 - 3 Under uncertainty about future output
- Empirical evidence from financial crisis of 2007 to 2010

Model

- Three dates: $t = 0, 1, 2$
- Sectors: Financial, Corporate, and Government

Financial sector:

$$\max_{s_0^s} E_0 \left[\left(w_s s_0^s - L_1 + \tilde{A}_1 + A_G + T_0 \right) \times 1_{\{-L_1 + \tilde{A}_1 + A_G + T_0 > 0\}} \right] - c(s_0^s)$$

- 1 Produces financial services s_0^s for per-unit wage w_s at cost of $c(s_0^s)$
 - an input to corporate sector production
 - revenue captured only if solvent at $t=1$ (otherwise goes to debtholders)
- 2 Incentive to produce depends on $p_{solv} = E_0 \left[1_{\{-L_1 + \tilde{A}_1 + A_G + T_0 > 0\}} \right]$
 - L_1 are liabilities due at $t=1$
 - $\tilde{A}_1$ uncertain payoff of assets at $t=1$
 - A_G a fraction k_A of outstanding sovereign debt
 - crisis $\rightarrow$ low p_{solv} (debt-overhang) $\rightarrow$ under-provision of financial services
 - T_0 is value of govt transfer (bailout)

Corporate Sector

Corporate sector:

$$\max_{s_0^d, K_1} E_0 \left[f(K_0, s_0^d) - w_s s_0^d + (1 - \theta_0) \tilde{V}(K_1) - (K_1 - K_0) \right]$$

- ① Buys s_0^d financial services to produce output $f(K_0, s_0^d)$ at $t=1$
- ② Makes investment K_1 at $t=1$ in project with uncertain payoff $\tilde{V}(K_1)$ at $t=2$
 - $V(K_1) = E_0 [\tilde{V}(K_1)] = K_1^\gamma, 0 < \gamma < 1$
- ③ Tax rate θ_0 set at $t = 0$ and levied at $t = 2$
 - funds existing govt debt and new transfer T_0
 - distorts incentive to invest $\rightarrow$ underinvestment:

$$\frac{dK_1}{d\theta_0} = \frac{V'(K_1)}{(1 - \theta_0)V''(K_1)} < 0$$

Example: HP threatens to reduce investment in Ireland if taxes hiked to fund bailout (11/21)

- **expected tax revenue** $\mathcal{T} = \theta_0 V(K_1)$
- $\mathcal{T}$ rises in θ_0 then falls (Laffer curve)

The Government's Problem

- ① Risk-Neutral representative consumer owns bonds and equity

⇒ Government's objective is to maximize expected total output

Uses Transfer (Bailout) to alleviate under-provision of financial services (debt-overhang)

- ② Funds the Transfer and Existing Govt Debt with Taxes:

- Existing Debt: N_D outstanding bonds with face value 1
- Transfer: N_T new bonds issued $\rightarrow T_0 = P_0 N_T$
- Defaults if: $\theta_0 \tilde{V}(K_1) < N_D + N_T \Rightarrow$ deadweight loss of D

- ③ Govt chooses tax rate θ_0 and new bond issuance N_T to maximize total output:

- subject to equilibrium conditions and price P_0
- **Insolvency ratio** $H = \frac{N_T + N_D}{\mathcal{T}} = \frac{N_T + N_D}{\theta_0 V(K_1)}$
- rewrite using $\mathcal{T}$ and H instead of θ_0 and N_T

Optimal $\mathcal{T}$ under Certainty and No Sovereign Default

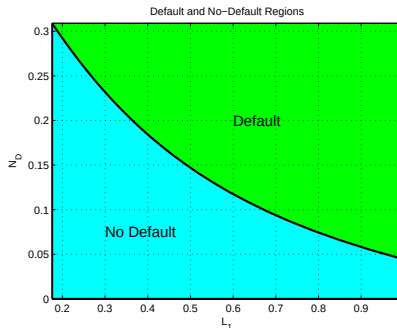
Consider first certain output, $\tilde{V}(K_1) = V(K_1)$, and no default $H = 1$

We show that under sufficient conditions:

- ① As $L_1 \uparrow$ (more severe debt-overhang) $\Rightarrow \hat{\mathcal{T}}$ (tax revenue) $\uparrow$ and $\hat{\mathcal{T}}_0$ (transfer) $\uparrow$
 - more severe debt-overhang $\rightarrow s_0 \downarrow$ (worse under-provision of financial services)
 - $\rightarrow$ greater marginal gain from increasing the transfer (bailout)
- ② As $N_D \uparrow$ (larger existing govt debt) $\Rightarrow \hat{\mathcal{T}}$ (tax revenue) $\uparrow$ but $\hat{\mathcal{T}}_0$ (transfer) $\downarrow$
 - larger debt $\rightarrow$ need more tax revenue to get same transfer ($\hat{\mathcal{T}} \uparrow$)
 - but incur greater underinvestment cost for same amount of transfer ($\hat{\mathcal{T}}_0 \downarrow$)

Strategic Sovereign Default Under Certainty

- ❶ Under strategic default, optimal to set $N_T \rightarrow \infty$ ($H \rightarrow \infty$)
- ❷ Captures full tax revenue by diluting existing bondholders to zero
 $\Rightarrow$ greater T_0 ($\uparrow s_0$) with lower θ_0 ($\downarrow$ underinvestment)
- ❸ But suffer dead-weight loss D



- k_A (fin sector sovereign holdings) $[-] \rightarrow$ 'collateral damage'

With Uncertainty

Uncertain output: $\tilde{V}(K_1) = V(K_1)\tilde{R}_V$

$$① \quad p_{def} = \text{prob}(\tilde{R}_V < H)$$

$$② \quad P_0 = E_0 \left[\min \left(1, \frac{1}{H} \tilde{R}_V \right) \right]$$

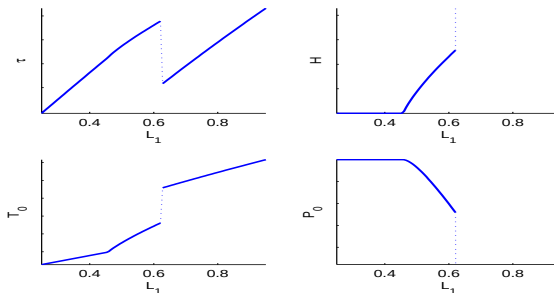
$$③ \quad T_0 = \left(\mathcal{T} - \frac{N_D}{H} \right) E_0 \left[\min(H, \tilde{R}_V) \right]$$

- Sovereign chooses H (insolvency ratio) on an interval, not just 1 or ∞

↑ $H \Rightarrow$ sovereign 'sacrificing' its creditworthiness to increase the bailout

- T_0 (bailout) ↑
- p_{def} (probability of sovereign default) ↑
- P_0 (govt bond price) ↓

Comparative Statics for Debt Overhang (L_1)



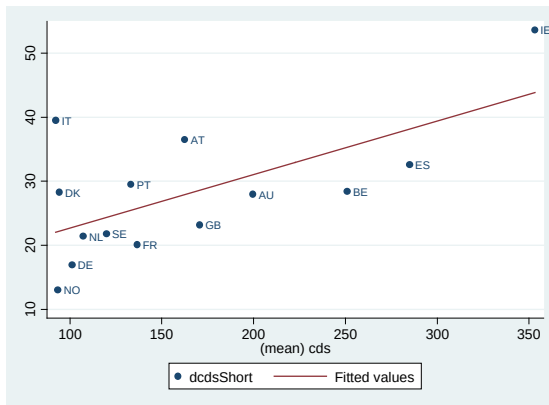
- T (expected tax revenue) increases in L_1
- High L_1 ('crisis') $\rightarrow H \uparrow$ (spillover, emergence of sovereign credit risk)
- $H \uparrow$ increases T_0 while $P_0 \downarrow$
 - dotted line shows when total default becomes optimal
 - default allows for larger T_0 with smaller T

Financial Sector → Sovereign

Fin sector crisis → severe debt-overhang (L_1) → Bailouts

- ① Bailouts reduce bank credit risk, trigger increase in sovereign credit risk
- ② *Spillover*: Pre-bailout financial sector distress predicts post-bailout increase in H (insolvency ratio) and sovereign CDS
- ③ Emergence of a positive relationship between the level of govt debt and sovereign credit risk (CDS)

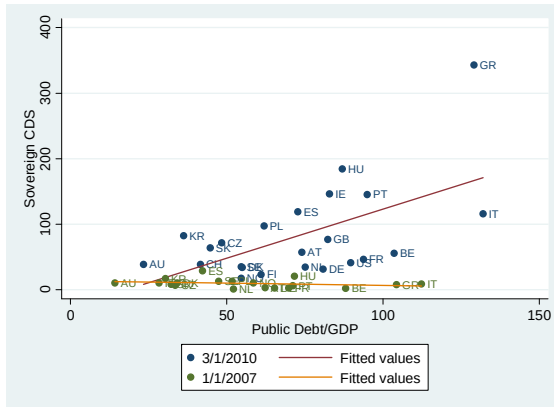
Spillover



Sov. CDS change vs. Pre-bailout Financial Sector Distress

- Financial Sector Distress: average bank CDS pre-bailout (21 Sep 2008)
- Sovereign CDS change: pre- to post-bailout

Emergence of Sovereign Credit Risk



Sov. CDS vs. Debt/GDP

- Pre-Bailouts: low- H region, not much relationship
- Post-Bailouts: sovereigns increase H , relationship becomes apparent

Spillover and the Emergence of Sovereign Risk

	Log (Sovereign CDS)			
	Pre-Bailout (1)	Post-Bailout (2)	Pre-Bailout (3)	Post-Bailout (4)
Pre-bailout Gov't Debt (in %)	0.006 (0.004)	0.005 (0.005)	0.015* (0.006)	0.013+ (0.007)
Pre-bailout Fin. Sector Distress		0.311 (0.208)		0.965* (0.357)
Observations	15	14	17	15
R-squared	0.134	0.171	0.261	0.488

Pre-bailout debt-to-gdp and fin sector distress

- *strongly predict* post-bailout sovereign CDS, debt-to-gdp
- no relation pre-bailouts

Sovereign → Financial Sector

Bailouts → emergence sovereign credit risk → affects bank credit risk

- 1 Increase in sovereign CDS raises Bank CDS
- 2 Empirical identification problem: unobserved third factor (e.g., gdp growth)
- 3 Examine co-movement of sovereign and bank CDS

$$\Delta \log(\text{Bank CDS}_{ijt}) = \alpha_i + \delta_t + \beta \Delta \log(\text{Sovereign CDS}_{jt}) + \gamma \Delta X_{ijt} + \varepsilon_{ijt}$$

X_{ij} control for

- Market-wide factors
- Time and bank fixed-effects
- Bank stock return

Market-Wide Controls and Time Fixed-Effects

	$\Delta \text{ Log(Bank CDS)}$					
	Pre-Bailout		Bailout		Post-Bailout	
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \text{ Log(Sovereign CDS)}$	0.017 (0.010)	0.003 (0.017)	0.448* (0.169)	-1.293** (0.387)	0.221** (0.026)	0.163** (0.033)
$\Delta \text{ Log(CDS Market Index)}$	0.962** (0.043)		0.893** (0.216)		0.722** (0.034)	
$\Delta \text{ Volatility Index}$	0.671** (0.113)		-0.946** (0.238)		0.057 (0.051)	
Week FE	N	Y	N	Y	N	Y
Interactions	N	Y	N	Y	N	Y
Observations	2,891	2,891	254	254	6,500	6,500
Banks	62	62	53	53	59	59
R-squared	0.262	0.476	0.114	0.599	0.338	0.479

- post-bailout: β is positive, very statistically significant
- around bailouts: β negative

Controlling Also For Bank Stock Returns

	$\Delta \text{ Log(Bank CDS)}$					
	Pre-Bailout		Bailout		Post-Bailout	
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \text{ Log(Sovereign CDS)}$	0.014 (0.010)	0.004 (0.018)	0.449** (0.164)	-1.02 (1.034)	0.197** (0.028)	0.146** (0.033)
Equity Return	-0.306* (0.142)		-0.194 (0.185)		-0.145** (0.030)	
Other Controls	Y	Y	Y	Y	Y	Y
Week FE	N	Y	N	Y	N	Y
Interactions	N	Y	N	Y	N	Y
Observations	2,891	2,891	254	254	6,500	6,500
Banks	62	62	53	53	59	59
R-squared	0.271	0.517	0.126	0.854	0.349	0.495

- sovereign CDS *still* very significant
- govt guarantees favor debt over equity → change in value of guarantee matters *even* after controlling for stock return

Conclusion

- Future costs of bailouts (e.g., moral hazard) are far from being the only important ones
- Costs are clear and present as bailouts have led to the emergence of sovereign credit risk
 - Gov. Budget constraint has tightened (gov. pockets are finite)– the elimination of slack is priced by the markets
- Resulting credit riskiness of sovereign debt feeds back onto financial sector
 - the ongoing banking crisis: impact of default risk in Greece, Ireland, Portugal, Italy
- Immediate stabilization of the financial sector by bailouts can be a Pyrrhic victory
 - the restructuring of financial sector debt should be considered more seriously

What if the Sovereign Cannot Do a Bailout? – Iceland vs. Ireland CDS

